Supplementary material

Table S1. Guarantee by data assumption

Each cell gives a representative method, the assumption it needs, and the price it pays. “Open” means no method in the reviewed set covers that cell.

guarantee	exchangeable	covariate shift	non-exchangeable / temporal	known groups
marginal coverage	Vovk et al. (2005), Lei et al. (2018), Angelopoulos & Bates (2023). One calibration pass.	Tibshirani et al. (2019). Likelihood ratio known or estimable.	Barber et al. (2023), Oliveira et al. (2022). Gap bounded by the violation.	Vovk (2012). Per-group quantiles; small n per group.
group-conditional coverage	Vovk (2012), Gibbs et al. (2025). Group-wise quantiles; smaller effective n.	open	open	Vovk (2012), Gibbs et al. (2025). Group labels known at test.
pointwise conditional coverage	Barber et al. (2021), Hore & Barber (2025). Larger sets.	open	open	open
risk control	Bates et al. (2021), Angelopoulos et al. (2022), Angelopoulos et al. (2025). Monotone loss; grid and multiplicity.	open in the reviewed set	Farinhas et al. (2023). Bounded non-exchangeability.	Combes et al. (2026).

Table S2. Coverage against four surveys

Authors’ assessment of topical coverage, not of quality. A circle means the topic is not a focus of that survey, not that the survey is deficient. Symbols: full circle = comprehensive, half circle = partial, open circle = not a focus.

dimension	this paper	Angelopoulos & Bates 2021	Fontana et al. 2023	Campos et al. 2024	Zhou et al. 2025
foundations	half	full	full	half	full
scores	full	full	half	full	full
conditional coverage	full	half	full	half	half
risk control	full	half	half	half	half
non-exchangeable	full	open	half	half	half
language models	half	open	open	full	half
implementation audit	full	open	open	open	open
two-dataset experiments	full	open	open	open	open

The last two rows are the reason this paper exists. The first six rows are the reason it has to cite the other four.

Table S3. Experimental grid

dataset	train pool	strengths	calibration	test	model	scores	alpha	seeds
digits	898	44 / 224 / 898	449 for A and C; 100 / 225 / 449 for B	450	standardiser + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5
CIFAR-10	30,000	100 / 1,000 / 5,000 / 30,000	2,000 for A; 500 / 2,000 / 10,000 for B; 2,000 for C	10,000	ResNet-18 features + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5 for A and B; 3 for C
covertype	120,000-row subset	6,000 / 60,000	20,000	40,000	standardiser + logistic regression	naive, APS randomized, APS deterministic	0.05 / 0.10 / 0.20	5

The calibration column gives the sizes used in each audit. The strength column gives the number of training points used to fit the model; the calibration and test sets are disjoint from it.